

HyungGyu Oh

Real-Time 3D Engineer · Rendering & Performance Optimization

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PROFILE

Real-time 3D and simulation engineer with 10+ years building, optimizing, and debugging interactive 3D applications on constrained hardware. Currently a partner-facing software development consultant at Unity, working inside customer projects to reproduce and resolve issues: profiling CPU, GPU, and memory on PC and mobile down to the draw call and the individual allocation. Before Unity, built a synthetic-data generation tool for autonomous-driving AI training and shipped a self-published Unity mobile title end to end, including a procedural terrain system built from real-world geospatial data and server-authoritative networking. B.S. in electrical engineering and M.S. in mechanical engineering with a digital image processing focus, including graduate work in C/C++ real-time signal processing on an automotive FMCW radar prototype.

EXPERIENCE

- Unity Technologies – Senior Software Development Consultant** Nov 2024 – Present · Seoul
- Diagnose and resolve **CPU/GPU, memory, and rendering bottlenecks** in real-time 3D applications through low-level profiling (Perfetto, RenderDoc, Xcode Instruments), translating capture data into actionable engineering fixes.
 - Take ownership of optimization work directly in client projects, from profiling through to the code change.
 - Serve as the **primary technical advisor** for a portfolio of game studios, building trusted, long-term relationships and supporting partners throughout the development lifecycle.
 - Extend Unity's Memory Profiler and CPU Profiler using AI coding agents: CSV export of untracked native allocations and cross-frame GC.Alloc aggregation, turning repeated manual diagnostics from live engagements into reusable tooling.
- Unity Technologies – Senior Technical Trainer** Aug 2022 – Oct 2024 · Seoul
- Delivered instructor-led engineering training on **profiling and optimization, URP and HDRP rendering pipelines, asset memory management, and shaders** to technical teams across multiple industries.
- REH Inc. – Founder / Lead Engineer** Feb 2014 – Aug 2022 · Seoul
- Founded and ran an independent studio, leading development of **Force Masters** from concept to launch on iOS and Android; owned engineering, platform compliance, and the store-review cycle on both platforms.
- Zynga – Software Engineer, Words with Friends (iOS)** 2013 · SF Bay Area, CA
- Shipped features on a top-grossing mobile franchise (Objective-C): IAP server sync, player-to-player messaging.
- NASA Jet Propulsion Laboratory – Software Engineer Intern** 2012 · Pasadena, CA
- Built interactive Mars-outreach experiences in Unity3D: custom shaders and C# performance optimization.
- Sogang University / Hanyang University – Lecturer / Adjunct Professor** 2014 – 2022 · Seoul
- Taught Unity and mobile application development across 5+ years of semesters while running the studio.

SELECTED PROJECTS

- Force Masters: Location-Based Mixed-Reality Mobile Game** · Unity · *Best Immersive Top 3, Made with Unity Korea 2021*
- Sole engineer on both the client and the game server. Key systems built:
 - On-demand voxel terrain generation from public geospatial data
 - Location-based matchmaking
 - Server-authoritative online gameplay
- Synthetic Image Generation for Self-Driving AI Training** · Unity · C#
- Developed a vehicle-data simulation tool generating driver-POV video/images while varying weather, time of day, road signs, and traffic.
- FMCW Automotive Radar Signal-Processing Module (Prototype)** · C++ (MFC) · DSP · DAQ

- Built the A/D converter controller and radar signal-processing pipeline end to end on a real-time multithreaded architecture with DAQ hardware control and a 50 ms view-update loop.

CORE SKILLS

Real-time 3D development (Unity): Scriptable Render Pipeline (URP in production, HDRP through instructor-led training), Shader Graph and HLSL shader authoring, uGUI, Addressables and asset pipeline, lighting, post-processing, 3D math (vectors, quaternions, matrices)

Profiling & optimization: CPU/GPU time, memory footprint, draw calls & batching, thermal behavior · RenderDoc, Perfetto, Android Studio, Xcode Instruments, Unity's own Profilers

AI-assisted engineering: AI coding agents (Claude Code) for development, debugging, and customer issue reproduction · building internal tooling from recurring consulting workflows

Customer-facing delivery: Technical advisory, complex-issue reproduction/debugging, project reviews, workshops/training, collaboration with Unity's engineers across Korea and EMEA

Languages & platforms: C#, C/C++, Objective-C, Java, JavaScript, Python, SQL · Android, iOS, PC, WebGL · Git

EDUCATION

M.F.A., Interactive Media (Game Design) – University of Southern California, School of Cinematic Arts

M.S., Mechanical Engineering (Digital Image Processing) – Seoul National University

B.S., Electrical Engineering – Korea University

Additional: Inventor of 3 U.S. patents (location-based in-game advertising) · Languages: native Korean, fluent English